# MandiMitra — Final Technical Report

**Explainable mandi price intelligence** · Vishnu Keer, Vivekananda Global University, Jaipur · 28 October 2026

Repository https://github.com/vishnu-keer/mandimitra · Live app https://mandimitra.streamlit.app/

> **How to read the labels. [Measured]** = produced by the released code from the released data on 28 Oct 2026 (file named). **[Reported]** = a statement from documentation or an external source, cited. **[Estimate]** = a judgement, marked as such. **[Proposed]** = future work, not built.

## 1. Executive summary

MandiMitra evaluates 1–3-day-ahead forecasts of daily wholesale onion and tomato prices (official Department of Consumer Affairs series, Jaipur, 1 Jan 2009 – 28 Apr 2015) against "tomorrow = today", and presents the result in a bilingual dashboard.

- **[Measured]** The model chosen in advance by rolling-origin cross-validation never had a lower test RMSE than Naive: Onion chose Naive at all three horizons; Tomato's choices were 5.66%, 0.53% and 0.31% worse (`outputs/best_model.csv`).
- **[Measured]** Naive had the lowest test MAE at all six pair-horizons.
- **[Measured]** In 48 Diebold–Mariano comparisons (Holm-adjusted), no model was significantly more accurate than Naive (`outputs/experiments/dm_test.csv`).
- **[Measured]** The first version's "7.33% better than Naive" came from selecting on the test set; that gap has DM p = 0.505.
- **[Measured]** 83–85% of reported days repeat the previous price; 80% forecast ranges covered only 55–74% on test, so no band is drawn.
- **[Measured]** 154 automated tests pass; rebuilds are byte-identical on CPython 3.11/3.12; CI is green; the live app was checked in a browser.

The product's value is trustworthy context — a strong reference forecast, data freshness, stale-quote rates and honest uncertainty — not a leaderboard.

## 2. Problem and target users

Daily mandi prices are volatile (Jaipur onion: Rs. 600–900/q in June 2013, Rs. 4,500 on 21 Oct 2013 **[Measured]**). Price-prediction tools are promoted to farmers and traders; their accuracy claims are rarely auditable.

| User | Need | What MandiMitra gives |
|---|---|---|
| Farmer | Should I expect the price to move in the next few days? | The last reported price, a forecast that never claims more than it can show, and whether any mode |
| Trader | How noisy is this market? | Stale-quote rate, reporting gaps, volatility comparison |
| Student / researcher | Is this evaluation trustworthy? | Protocol, tests, hashes, significance, hindsight winners shown separately |
| Market analyst | Can I reuse the data? | Provenance, licence notes, CSV downloads |

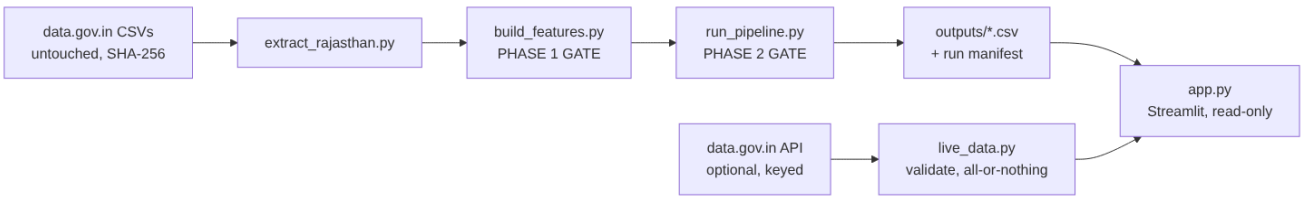
**[Reported]** No user research has been conducted; the needs above are assumptions to be tested (`docs/USER_RESEARCH_PLAN.md`).

## 3. Requirements

**Functional.** Commodity and market selection limited to validated series; historical trends with reported vs carried-forward days; market comparison; 1/2/3-day forecasts with origin and target dates; model comparison against Naive with both MAE and RMSE; actual-vs-predicted and residual charts; data quality and freshness; uncertainty findings; source attribution; CSV downloads; English/Hindi; current official prices when an API key is configured.

**Non-functional.** ₹0 hosting; no training on page load; reproducible outputs; no secrets in the repository; phone-width layout; light theme pinned for projectors; every number traceable to a file; failures shown as actionable messages, not tracebacks.

## 4. Architecture




Architecture: official files to read-only dashboard, with the optional current-price path

Modules and responsibilities: `docs/ARCHITECTURE.md`. Key decisions: files not a database (small data, ₹0, auditable in git); direct multi-horizon models (no recursive leakage); a dashboard that cannot import a model library (enforced by a test); gates that exit non-zero.

**Data flow.** Official CSV → Rajasthan subset (byte-identical extract) → clean (zeros → missing, range filter, de-dup) → daily calendar → features and observed-target masks → CV folds → selection → refit → single test scoring → forecast and intervals → CSVs → manifest → dashboard.

## 5. Data source and quality

**[Measured]** (`docs/DATA_REPORT.md`): Onion/Jaipur 1,288 reported prices of 1,551 rows (55.8% of calendar days), 1,002 usable 1-day rows; Tomato/Jaipur 1,228 (53.2%), 995. Jodhpur (315/322) and Kota (241/245) fail the 400-row gate. 35% of feature rows use a carried-forward input; no target is ever carried forward. 84.8% (Onion) and 82.8% (Tomato) of reported days repeat the previous price. Longest gap between reports: 70 days (Onion/Jaipur).

**[Reported]** Licence: platform Terms of Use require acknowledgement and point to per-dataset licence metadata, which is absent; GODL-India is the platform licence per secondary sources; its page redirected to a sign-in (`docs/01_DATA_SOURCE.md`).

## 6. Feature engineering

Decision point = end of day *t*. Twelve features (price, lags 1/3/7, MAs 3/7, 7-day volatility, 1-day return, deviation from MA, days since report, weekday, month). Windows need ≥ 60% reported days. Gap fill ≤ 2 days, inputs only — **[Measured]** 391 / 1,002 / 1,145 usable rows for 1 / 2 / 3 days of fill. Leakage defences: backward windows (code-scanned), direct targets, truncate-and-compare proof, future-perturbation test, causal ARIMA filter, target-date embargo.

## 7. Models

Naive; ARIMA (AIC order, causal h-step Kalman forecasts); Random Forest (300 trees, depth 6, leaf 8); Gradient Boosting (200 stages, lr 0.05, depth 3, leaf 8, subsample 0.9); Random Forest-delta (predicts the change). Seed 42; single-threaded trees for bit-exact output.

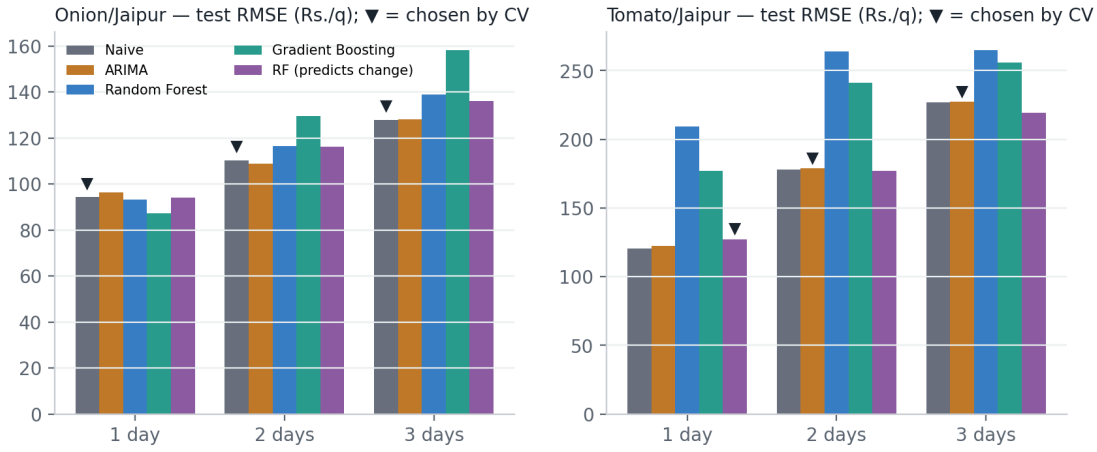
## 8. Evaluation methodology

Final test = last 20% of dates, scored once after selection. Selection = lowest mean fold RMSE over 4 expanding rolling-origin folds inside the development period (rule fixed in advance). Embargo on target dates. Identical rows per model. MAE and RMSE in Rs./quintal; MAPE in the files only; $R^2$ not shown. Diebold–Mariano (HLN) tests with Holm adjustment. Disclosed judgement calls: validation redesign after seeing validation results; fifth candidate admitted on CV; tree settings from the first protocol (audited — one selection changes under defaults, no conclusion changes).

## 9. Results

**Table 1 [Measured]. Test period, model chosen by CV (Rs./quintal).**

| Series | h | Chosen | Test RMSE | Naive RMSE | vs Naive | Test MAE | Naive MAE |
|---|---|---|---|---|---|---|---|
| Onion | 1 | Naive | 94.27 | 94.27 | 0.00% | 39.75 | 39.75 |
| Onion | 2 | Naive | 110.29 | 110.29 | 0.00% | 53.23 | 53.23 |
| Onion | 3 | Naive | 127.93 | 127.93 | 0.00% | 65.85 | 65.85 |
| Tomato | 1 | RF-delta | 127.33 | 120.51 | −5.66% | 70.92 | 54.27 |
| Tomato | 2 | ARIMA | 179.10 | 178.16 | −0.53% | 99.25 | 90.55 |
| Tomato | 3 | ARIMA | 227.38 | 226.68 | −0.31% | 131.68 | 123.77 |




Test RMSE of every candidate

**Table 2 [Measured]. Every candidate's test RMSE.**

| Series | h | Naive | ARIMA | RF | GBM | RF-delta |
|---|---|---|---|---|---|---|
| Onion | 1 | 94.27 | 96.34 | 93.30 | 87.26 | 93.94 |
| Onion | 2 | 110.29 | 108.94 | 116.58 | 129.44 | 116.34 |
| Onion | 3 | 127.93 | 128.19 | 138.81 | 158.14 | 136.17 |
| Tomato | 1 | 120.51 | 122.16 | 209.42 | 176.76 | 127.33 |
| Tomato | 2 | 178.16 | 179.10 | 263.80 | 241.13 | 176.91 |
| Tomato | 3 | 226.68 | 227.38 | 264.76 | 255.89 | 219.18 |

Lower-than-Naive cells exist (e.g. GBM Onion h=1) but none was chosen in advance and none is statistically distinguishable from Naive.

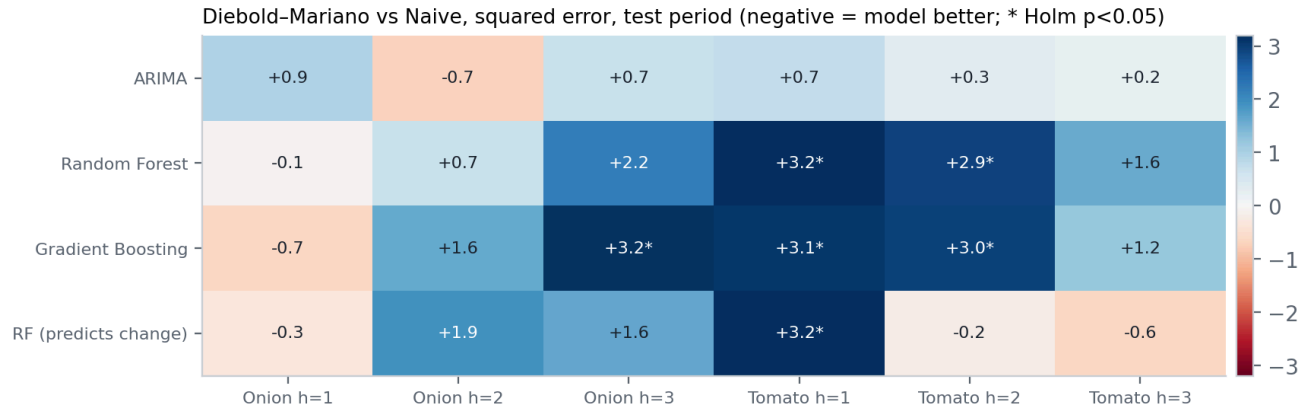
## 10. Baseline comparisons

**[Measured]** First version vs now (Onion/Jaipur):

| h | First version (picked on test) | Now (picked by CV) |
|---|---|---|
| 1 | Gradient Boosting, 87.36, “+7.33%” | Naive, 94.27 |
| 2 | ARIMA, 108.94, “+1.23%” | Naive, 110.29 |
| 3 | Naive, 127.93 | Naive, 127.93 |

**[Measured]** Weekly seasonal naive is far weaker than Naive (mean CV RMSE Onion 169.5/160.5/168.2 vs 63.0/84.9/117.3; Tomato 215.0/212.9/213.3 vs 79.4/109.7/136.5).

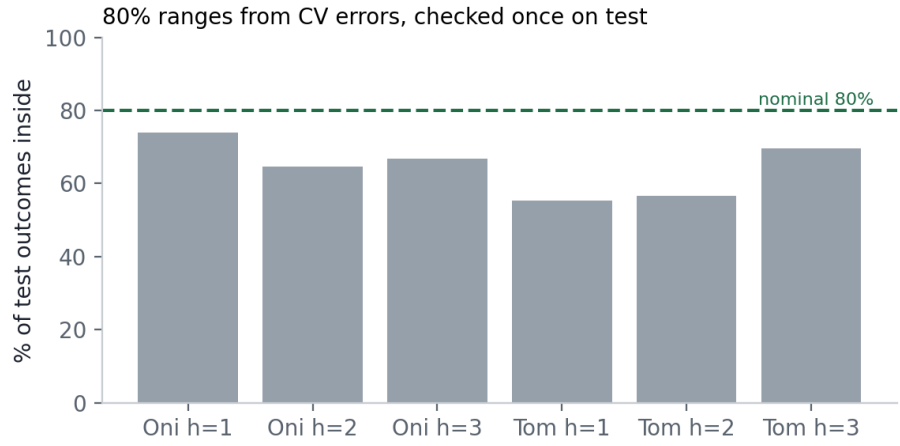
**[Measured]** Diebold–Mariano: squared loss 0 of 24 favour a model, 6 favour Naive; absolute loss 0 of 24 favour a model, 24 favour Naive.



Diebold–Mariano statistics

## 11. Forecast uncertainty

**[Measured]** 80% empirical ranges from CV errors covered 74.0 / 64.7 / 66.8% (Onion) and 55.3 / 56.7 / 69.6% (Tomato) of test outcomes. None held up; the dashboard shows this finding instead of a band.



Interval coverage

## 12. Testing

**[Measured]** `python -m pytest` → 154 passed (CPython 3.12, clean install).

| Area | File | Tests |
|---|---|---|
| Leakage | `test_features_no_leakage.py` | 9 |
| Chronology | `test_splits_chronology.py` | 10 |
| Protocol (CV, embargo, identical rows, ARIMA causality) | `test_evaluation_protocol.py` | 33 |
| Forecast validity | `test_forecast_validation.py` | 17 |
| Data validation | `test_data_validation.py` | 19 |
| Metrics | `test_metrics.py` | 10 |
| Provenance | `test_provenance.py` | 5 |
| Manifest + no training in app | `test_manifest.py` | 6 |
| Dashboard (headless AppTest) | `test_dashboard.py` | 15 |
| Current-price ingestion (validation, schema change, paging, failure recovery, key never leaked, staleness) | `test_live_data.py` | 25 |
| Diebold–Mariano | `test_significance.py` | 5 |

Browser checks: 33/33 local (Playwright), plus a manual-equivalent DOM check of the live app. Details: `docs/TEST_REPORT.md`.

## 13. Performance and deployment

**[Measured]** Pipeline 20–60 s; first page load 2–5 s locally (nothing is trained on load). Streamlit Community Cloud, Python 3.12 (the platform default 3.14 failed to build the pinned stack — documented in `docs/DEPLOY.md`). CI on GitHub Actions (3.11 + 3.12): tests, full rebuild, reproducibility check, server health.

## 14. Security, privacy and licensing

No accounts, cookies set by the app, analytics or personal data. The optional API key is read from an environment variable or Streamlit secret, never written, logged or shown (tested). `.gitignore` excludes `.env`, secrets, virtual environments and fetched snapshots. Code: MIT. Data: GODL-India with attribution (`DATA_LICENSE.md`); confirm the licence wording manually.

## 15. Limitations

One mandi, two commodities, 2009–2015, price history only; retrospective only; ~200 test days per horizon; stale quotes; disclosed researcher degrees of freedom; no user validation; current-data path untested against a real key.

## 16. Cost and sustainability

**[Measured]** ₹0 spent: GitHub (public repo, Actions), Streamlit Community Cloud, data.gov.in. **[Reported]** Community Cloud: free, public apps, up to 2.7 GB RAM, sleeps after 12 h idle; free tiers can change (`docs/BUDGET_AND_ROADMAP.md`).

## 17. Roadmap [Proposed]

Current data (API key; evaluate the CEDA daily mirror's terms); more markets through the gate; arrivals and rainfall inputs; prospective scoring of live forecasts; consented user study; mobile-first Hindi view. Week-by-week plan: `docs/BUDGET_AND_ROADMAP.md`.

## 18. Reproduction

```
git clone https://github.com/vishnu-keer/mandimitra.git && cd mandimitra
python3.12 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python scripts/extract_rajasthan.py
python scripts/build_features.py
python scripts/run_pipeline.py --all-pairs
python scripts/experiment_significance.py
python -m pytest
streamlit run app.py
```

## Appendix A — files

`outputs/best_model.csv`, `model_comparison.csv`, `predictions.csv`, `cv_folds.csv`, `forecast.csv`, `split_boundaries.csv`, `excluded.csv`, `feature_importance.csv`, `experiments/*.csv`, `legacy/*.csv`, `run_manifest.json`.

## Appendix B — references

See `research/MANDIMITRA_RESEARCH_PAPER.md` (references verified against Crossref or publisher pages; notes in `research/references.bib`).